(Mostly Doing today than theory)

Python - 6 th Day (Important theory for function)

Topic - functions, while loops & Concepts

Main Goal

On Day 6 of the Python Bootcamp, you focus on building logic using functions and while loops by guiding a robot (in Reborg's world) to Escape a maze. This reinforces writing reusable code and decision - Based looping

Core Concepts Covered

(1) Python functions

functions let you group into a reusable block with a name

Syntax

```
def function_name():
    # Code here
```

- you call a function by using its name
- functions help organize logic (eg:- turning the robot)

Example

```
def right_turn():
    turn_left()
    turn_left()
    turn_left()
```

This defines right_turn() by turning left three times to simulate a right turn.

(2) while loops

while loops repeat instructions as long as a conditions remains true

- Used to keep moving the robot until it reaches a specific position
- Combine logic checks (if, elif, else) inside loops.

Example

```
while front_is_clear():
        move()
```

This keeps moving forward while the path ahead has no obstacles.

(3) Maze-solving logic (project: "Reborgs world")

Objective

Guide the robot from start to the goal by writing automated movement logic rather than one-step commands

Key control flow pattern

```
while not at_goal():
    if right_is_clear():
        right turn()
        move()
    elif front_is_clear():
        move()
    else:
        turn left()
```

Explanation:
- Repeat this cycle until the robot reaches the goal
- check if right is clear → turn right & move
- Else if front is clear → move forward
- Else → turn left
- this approach Enables the robot to follow walls or paths in the maze

(4) Combining loops & functions
- loops Control ongoing movement
- functions handle repeated Sub-tasks
- this Style makes your Code cleaner and Easier to debug.

(*) Skills you practice
- Defining and calling functions
- Understanding loops and truth Conditions
- Making decisions using if/elif/else
- logical problem solving inside Code
- Structuring Code in readable blocks
- Debugging logic when robot doesn't behave as Expected

Why this Matters
Day 6 bridges beginner Syntax (variable, loops) with automated logic, a Critical Step toward real problems like automation, Simulation and game logic